

Analyzing the Impact of Vaccination on the Spread and Control of COVID-19 in Israel

A Data Science and Mathematical Modeling Approach

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Background and Objectives

The COVID-19 pandemic has had a profound impact on global health, economies, and daily life. In late 2019, it became the first major pandemic the world has faced in decades. Governments and health organizations worldwide have implemented various measures to control its spread, including lockdowns, social distancing, and, most importantly, vaccination campaigns, which have been critical in reducing severe cases, hospitalizations, and deaths.

This project focuses on the spread and control of COVID-19 in Israel, one of the first countries to launch a nationwide vaccination campaign. On December 20, 2020, Israel launched a national COVID-19 vaccination campaign using BNT162b2 vaccines, administered through its four health maintenance organizations (HMOs), according to a prioritization schedule determined by the Israeli Ministry of Health [4]. Using real-world data provided by Our World in Data, this study aims to explore the dynamics of the pandemic in Israel by analyzing trends in infections, deaths, and vaccination progress. A mathematical model, specifically the SIRVD model (Susceptible-Infected-Recovered-Vaccinated-Dead), will be used to understand how vaccination efforts have influenced the spread of the virus.

Understanding these dynamics is not only of scientific interest, but also critical for guiding public health strategies and policymaking. By analyzing historical data and fitting it to mathematical models, this study provides insights into the effectiveness of interventions and the future trajectory of the pandemic.

Data Preparations

Dataset Description

The dataset used in this project is provided by Our World in Data, a widely used source for global COVID-19 statistics. It originally includes worldwide data, but for this study, we filtered it specifically for Israel, spanning from January 2020 to December 2021, covering both pre-vaccine and post-vaccine pandemic phases.

To analyze the impact of vaccination, we split the dataset into two separate periods:

- Period 1 (2020): January 2020 – December 2020 (before vaccination rollout).
- Period 2 (2021): January 2021 – December 2021 (after vaccination rollout began on December 20, 2020).

The key columns in the dataset are:

- **date**: The specific date of the observation.
- **new_cases**: The number of newly reported COVID-19 cases on a given day.
- **new_deaths**: The number of newly reported COVID-19 deaths on a given day.
- **total_cases**: The cumulative number of confirmed COVID-19 cases up to the given date.

- **total_deaths**: The cumulative number of deaths attributed to COVID-19 up to the given date.
- **stringency_index**: A measure of government-imposed restrictions (e.g., lockdowns), scored from 0 to 100.
- **people_vaccinated**: The cumulative number of individuals who received at least one dose of a vaccine (only in Period 2).
- **people_fully_vaccinated**: The cumulative number of individuals who received all required doses of a vaccine (only in Period 2).
- **population**: The total population of Israel.

Stringency Index

The **COVID-19 Stringency Index** is a composite measure developed by the Oxford COVID-19 Government Response Tracker (OxCGRT) to assess the strictness of government policies in response to the pandemic. It is calculated by averaging the scores of nine policy indicators, each normalized to a value between 0 and 100. These indicators include:

- School closures
- Workplace closures
- Cancellation of public events
- Restrictions on public gatherings
- Closures of public transport
- Stay-at-home requirements
- Public information campaigns
- Restrictions on internal movements
- International travel controls

Each indicator is assigned a score based on the level of restriction implemented, and the Stringency Index is the simple average of these scores. This results in a daily index value ranging from **0 to 100**, with higher values indicating more stringent government responses [2].

For a detailed explanation of the calculation methodology can be observed at OxCGRT's official documentation [1].

Levels of the Stringency Index:

- **0-20**: Minimal restrictions, mostly recommendations rather than enforced measures.
- **20-50**: Moderate restrictions, including partial lockdowns, public event cancellations, and workplace limitations.

- **50-80:** High restrictions, with stay-at-home orders, public transport closures, and limits on internal movement.
- **80-100:** Very strict restrictions, including full lockdowns, curfews, and severe travel restrictions.

Preprocessing and Data Aggregation

To ensure consistency, the dataset was preprocessed in several steps. First, the data was filtered to include only records from 2020 and 2021, excluding any data from 2022. Then the dataset was split into two separate parts: one covering 2020, before vaccines were introduced and another for 2021, after the mass vaccination began.

For cumulative columns such as the `people_vaccinated` and `people_fully_vaccinated`, missing values were filled using forward-filling. Additionally, before the vaccination campaign started, any missing vaccination values were set to zero.

Datasets were also converted from daily to weekly format. Weekly totals were calculated for `new_cases` and `new_deaths`, while cumulative values, including `total_cases`, `total_deaths`, and vaccination counts, were recorded as the last available value of each week. For `stringency_index`, a measure of government restrictions, weekly averages were used instead of daily values.

All preprocessing steps were performed using Python, specifically with the `pandas` and `numpy` libraries.

Descriptive Statistics and Visualizations

To gain insights into the trends of COVID-19 cases, deaths, and government interventions, we analyzed key descriptive statistics separately for 2020 (before vaccination) and 2021 (after vaccination rollout). Below are the key statistics for each year.

Metrics Summary

Table 1: Summary Statistics for 2020 and 2021

Metric	2020	2021
Average Weekly Cases	7,904.36	18,835.69
Average Weekly Deaths	66.84	95.53
Average Stringency Index	63.07	52.83
Average Vaccination Rate (%)	0.00	54.86
Average Fully Vaccinated Rate (%)	0.00	48.10

The results show a sharp increase in average weekly cases in 2021 compared to 2020. However, despite higher case numbers, the average weekly deaths only moderately increased. The average stringency index was higher in 2020, reflecting stricter lockdown measures before vaccines were introduced. By 2021, vaccination rates had reached over 50% of the population, which played a key role in reducing severe outcomes.

Trends Over Time

Weekly New Cases and Deaths

The figures below depict the progression of new COVID-19 cases and deaths on a weekly basis. Deaths are scaled x100 for clarity.

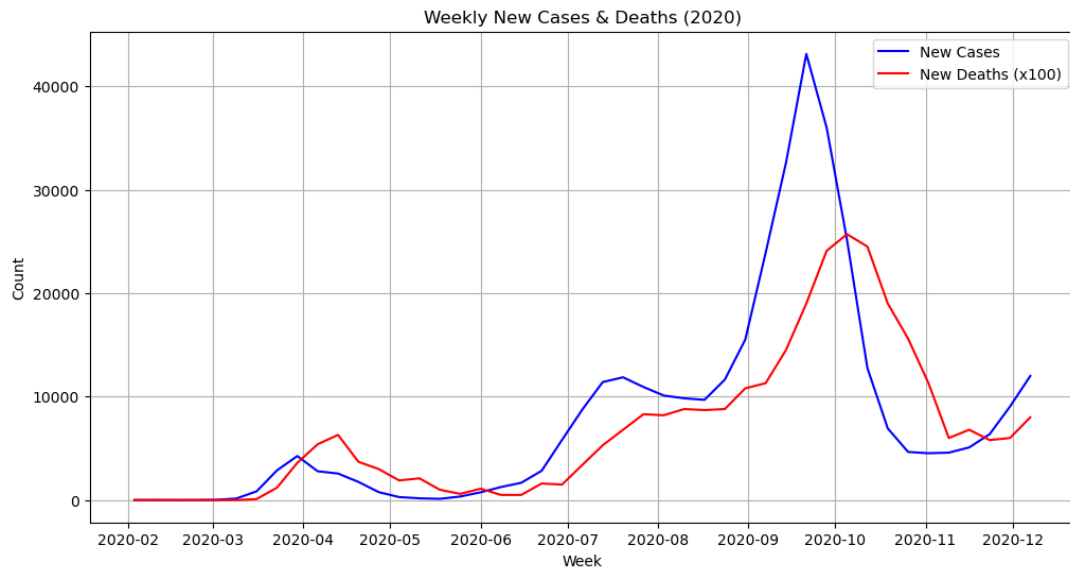


Figure 1: Weekly New Cases and Deaths in 2020

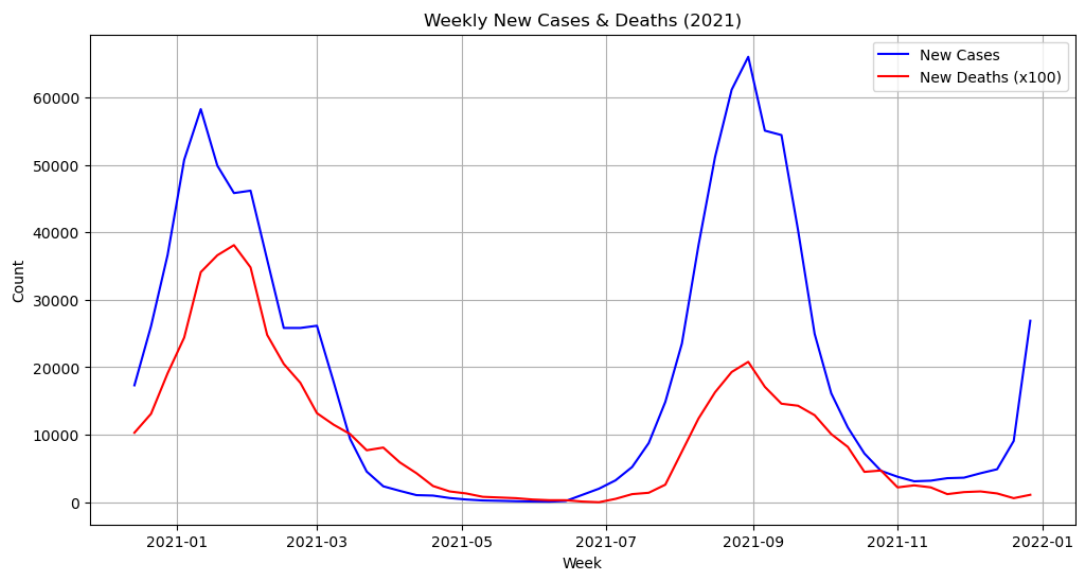


Figure 2: Weekly New Cases and Deaths in 2021

The graphs show multiple waves of infections, with cases and deaths peaking in late 2020 and during major waves in 2021. Despite higher case numbers in 2021, the overall mortality remained relatively stable, likely due to the impact of vaccinations.

Government Stringency Measures

To assess the severity of government interventions, we analyze the stringency index, which measures the strictness of policies such as lockdowns and movement restrictions.

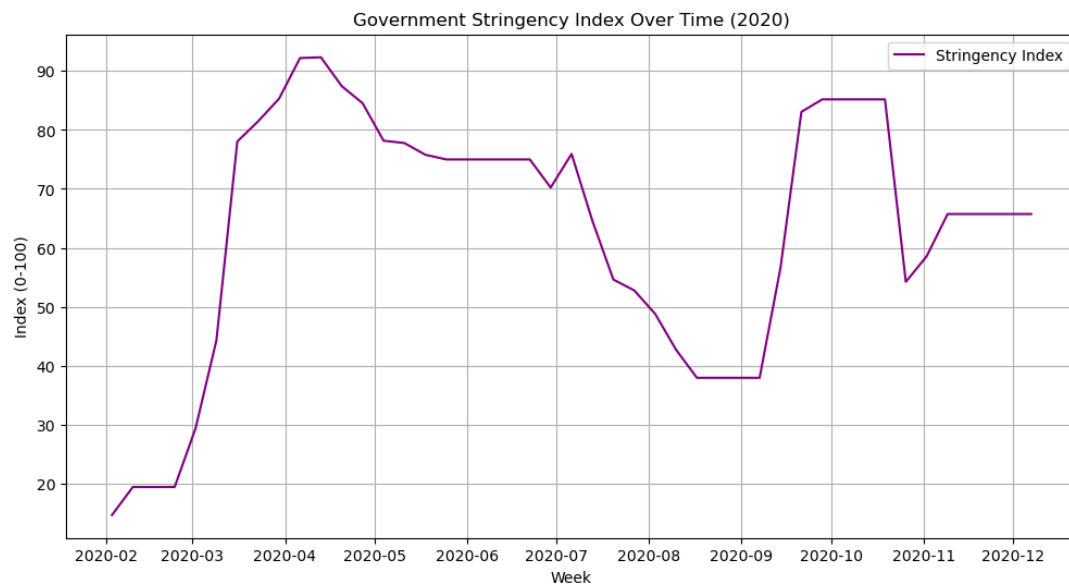


Figure 3: Government Stringency Index in 2020

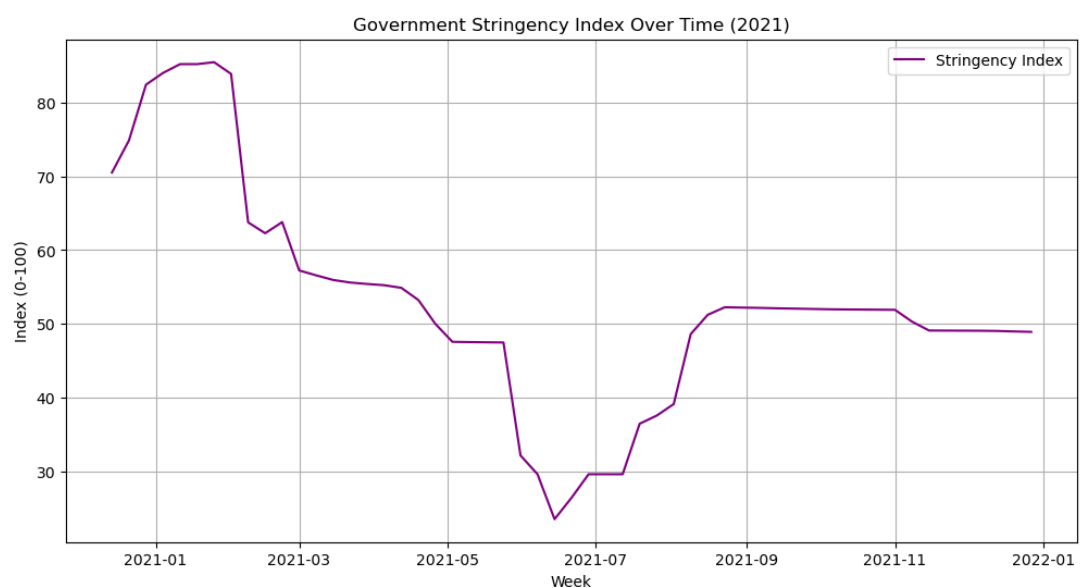


Figure 4: Government Stringency Index in 2021

Stringency levels were at their highest in early 2020, reflecting strict early containment measures. In contrast, 2021 shows a gradual decline in restrictions, coinciding with the vaccination campaign.

Vaccination Progression

The following graph illustrates the rapid rollout of vaccinations in Israel during 2021, showing the percentage of the population that received at least one dose and those who were fully vaccinated.

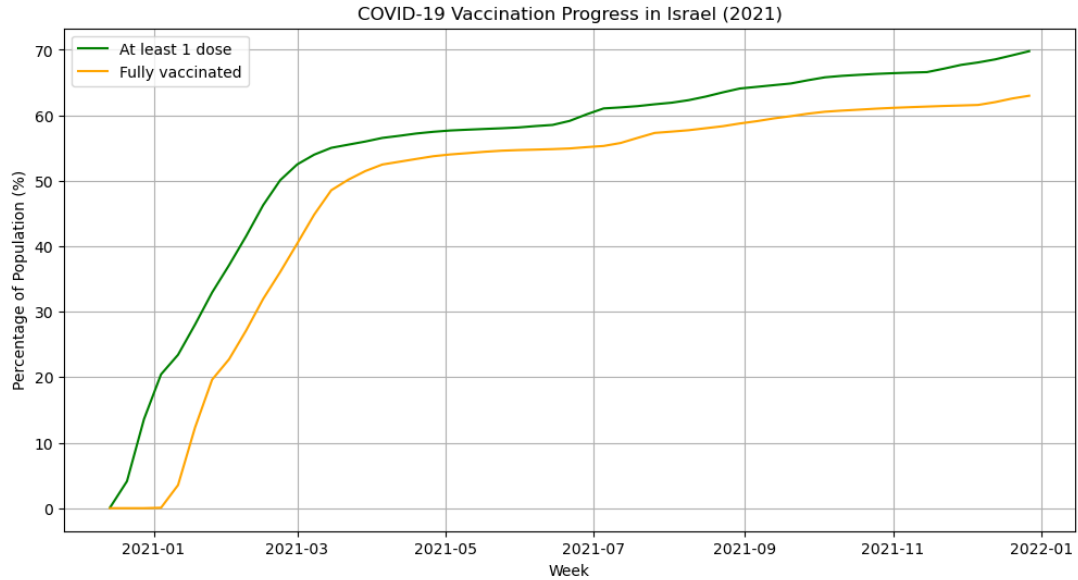


Figure 5: Vaccination Progress in Israel (2021)

By mid-2021, over 50% of the population had been fully vaccinated, significantly reducing severe illness and mortality.

Comparison of Averages Between 2020 and 2021

Lastly, we provide a visual comparison of key averages between 2020 and 2021.

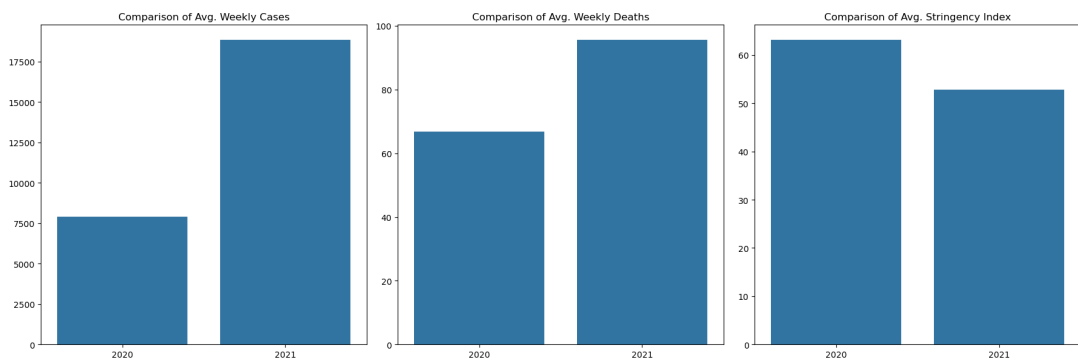


Figure 6: Comparison of Average Weekly Cases, Deaths, and Stringency Index Between 2020 and 2021

This comparison highlights the impact of vaccinations: despite a significant rise in cases in 2021, the number of deaths remained relatively controlled. The reduced stringency index also suggests a shift from strict lockdown measures to a reliance on vaccination as the primary control strategy.

Statistical Models

To gain deeper insights into the relationships between key variables in our dataset, we applied two statistical modeling approaches: correlation analysis and multiple linear regression (MLR). The correlation matrices allow us to explore associations between cases, deaths, and policy measures, while the regression models help us quantify the impact of different factors on COVID-19 deaths.

Correlation Analysis

We first examined the correlation between key variables using Pearson's correlation coefficient. The correlation matrices for 2020 (pre-vaccination) and 2021 (post-vaccination) are shown below.

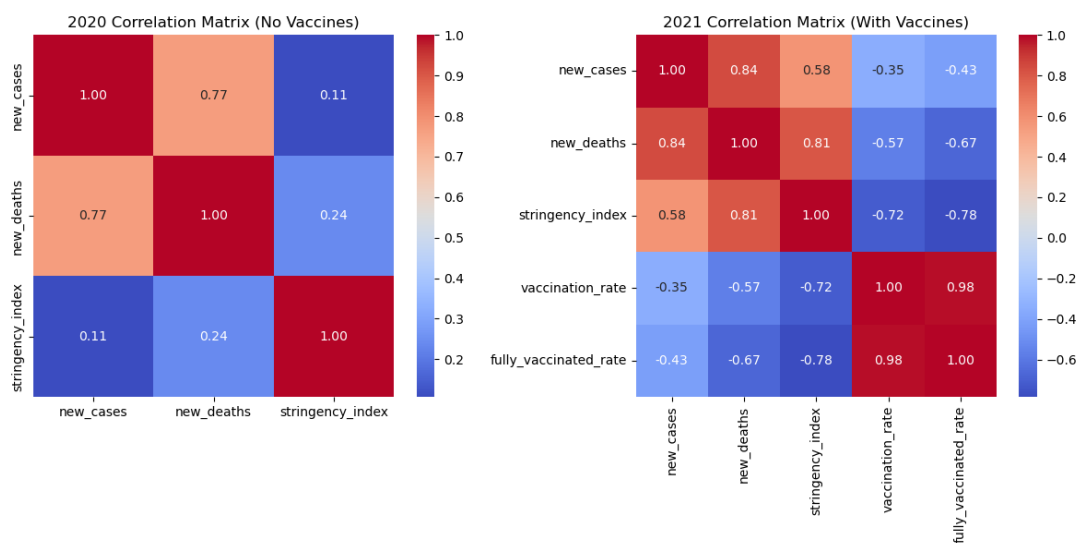


Figure 7: Correlation Matrices for 2020 (Left) and 2021 (Right)

- In 2020, before vaccines, new cases and deaths had a strong correlation (0.77), indicating that higher infections led to more fatalities.
- The stringency index had a weak correlation with cases and deaths, suggesting that restrictions alone were not enough to stop the pandemic.
- In 2021, the correlation between new cases and deaths remained strong (0.84), but restrictions showed a higher correlation with deaths (0.81), implying a greater role in reducing severe cases.
- Vaccination had a significant negative correlation with both cases (-0.35) and deaths (-0.57), showing that increased vaccine coverage had a stronger effect in lowering mortality than restrictions.

Multiple Linear Regression Analysis

To further investigate the impact of different variables on COVID-19 deaths, we conducted multiple linear regression (MLR) analysis for both 2020 and 2021.

MLR Results for 2020 (Pre-Vaccination)

Table 2: MLR Results for 2020 (Dependent Variable: New Deaths)

Variable	Coefficient
Intercept	24.18
New Cases	0.0054
R^2	0.591
Adjusted R^2	0.582
F-statistic	62.16

For 2020, the regression model shows that new cases were the primary predictor of new deaths, with a significant positive coefficient (0.0054, $p < 0.001$). The model explains approximately 59.1% of the variance in new deaths, suggesting a strong direct relationship between infections and mortality before vaccination efforts began.

MLR Results for 2021 (Post-Vaccination)

Table 3: MLR Results for 2021 (Dependent Variable: New Deaths)

Variable	Coefficient
Intercept	-80.19
New Cases	0.0026
Stringency Index	2.07
Fully Vaccinated Rate	-5.60
R^2	0.897
Adjusted R^2	0.889
F-statistic	108.6

In 2021, the regression model shows a more complex relationship between new deaths and multiple factors. While new cases remained a strong predictor of mortality (0.0026, $p < 0.001$), government restrictions and vaccination rates also played a significant role. The negative coefficient for fully vaccinated rates (-5.60, $p < 0.01$) suggests that higher vaccination coverage was associated with a decrease in deaths, even after controlling for new cases and government interventions. This model explains 89.7% of the variance in new deaths, highlighting the effectiveness of vaccination campaigns in reducing severe outcomes.

- In 2020, new cases were the strongest predictor of new deaths, with government interventions having a limited direct impact.
- By 2021, the introduction of vaccines significantly altered the dynamics, with fully vaccinated rates negatively correlating with deaths.
- The regression model for 2021 suggests that a combination of vaccinations and policy measures was essential in mitigating COVID-19 mortality.
- The higher explanatory power of the 2021 model ($R^2 = 0.897$) suggests that additional factors, such as immunity and medical advancements, contributed to lower fatalities despite rising case numbers.

Mathematical Model: SIRD and SIVRD

Model Description

To analyze the spread and control of COVID-19, we use two compartmental models:

- The **SIRD model** for 2020, which includes Susceptible (S), Infected (I), Recovered (R), and Deceased (D) populations.
- The **SIVRD model** for 2021, which extends SIRD by incorporating a Vaccinated (V) compartment to reflect the effects of vaccination.

The ordinary differential equations (ODEs) governing these models are:

SIRD model (2020, no vaccination):

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N} \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - (\gamma + \mu)I \\ \frac{dR}{dt} &= \gamma I \\ \frac{dD}{dt} &= \mu I\end{aligned}\tag{1}$$

SIVRD model (2021, including vaccination):

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N} - \nu S \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - (\gamma + \mu)I \\ \frac{dV}{dt} &= \nu S \\ \frac{dR}{dt} &= \gamma I \\ \frac{dD}{dt} &= \mu I\end{aligned}\tag{2}$$

where:

- β is the transmission rate.
- γ is the recovery rate.
- μ is the mortality rate.
- ν (only in 2021) is the vaccination rate.

Parameter Estimation

We estimated the parameters using a non-linear least squares approach, minimizing the residuals between the observed and predicted values. The optimization was performed separately for each model and epidemic wave.

To estimate each parameter:

- **Transmission rate (β):** Derived by fitting the equation governing infections to real case data.
- **Recovery rate (γ):** Estimated by optimizing the match between the model's recovered cases (R) and the expected recovery trends inferred from the model.
- **Mortality rate (μ):** Optimized by fitting the cumulative deaths to real-world data.
- **Vaccination rate (ν) (for 2021):** Determined by matching the model's predicted vaccinated population (V) to actual vaccination data.

The parameters were optimized using the trust-region reflective method with multiple random starting points to ensure robustness, using `scipy.optimize.least_squares`. The objective function minimized the weighted sum of squared residuals:

$$\min \sum \left(\frac{\hat{I} - I}{\sigma_I} \right)^2 + \sum \left(\frac{\hat{D} - D}{\sigma_D} \right)^2 + \sum \left(\frac{\hat{V} - V}{\sigma_V} \right)^2 \quad (3)$$

where:

- $\hat{I}, \hat{D}, \hat{V}$ are the predicted infections, deaths, and vaccinated individuals.
- $\sigma_I, \sigma_D, \sigma_V$ are the standard deviations of each variable, used for scaling.

Results of Parameter Optimization

For waves 1 and 3:

- **Optimized SIRD parameters** (2020, no vaccines): $\beta = 30.76$, $\gamma = 29.88$, $\mu = 0.0136$.
- **Optimized SIVRD parameters** (2021, with vaccines): $\beta = 1.182$, $\gamma = 0.873$, $\mu = 0.0065$, $\nu = 0.0567$.

For waves 2 and 4:

- **Optimized SIRD parameters** (2020, no vaccines): $\beta = 4.531$, $\gamma = 4.191$, $\mu = 0.0093$.
- **Optimized SIVRD parameters** (2021, with vaccines): $\beta = 6.755$, $\gamma = 2.028$, $\mu = 0.0034$, $\nu = 0.01196$.

Model Fit and Performance

The following plots compare the actual vs. predicted values for infections, deaths, and vaccinations:

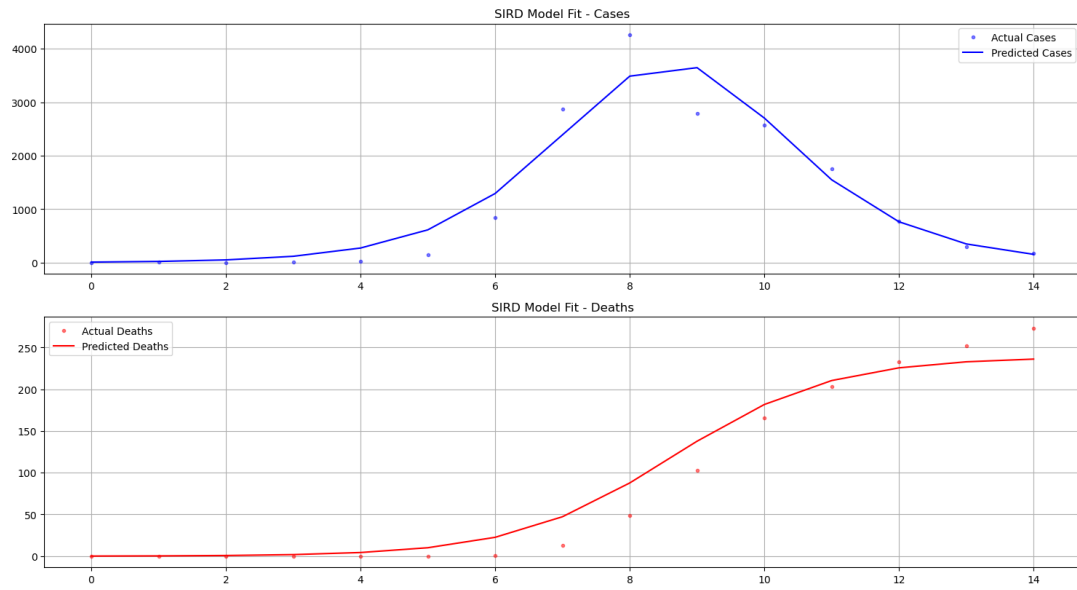


Figure 8: SIRD Model Fit for Wave 1 (2020)

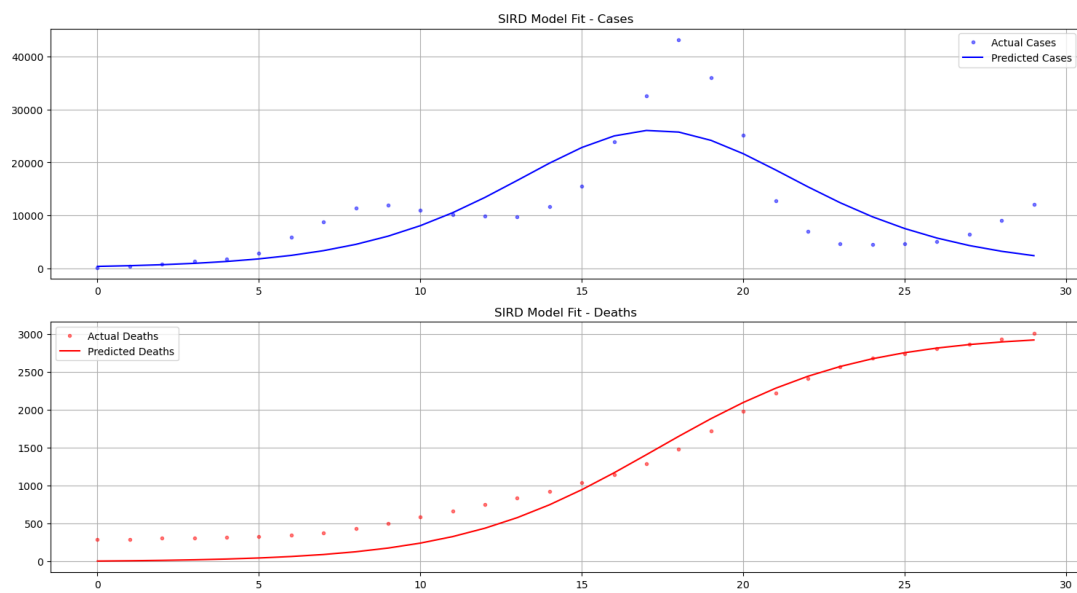


Figure 9: SIRD Model Fit for Wave 2 (2020)

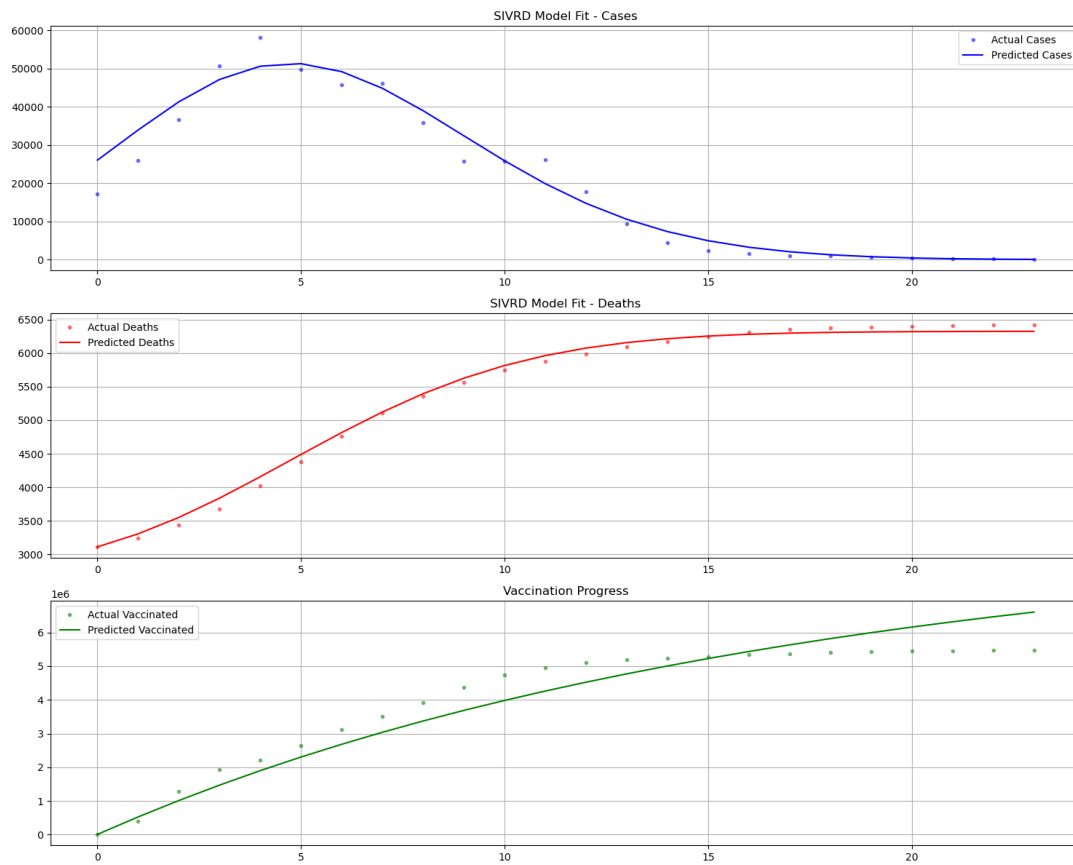


Figure 10: SIVRD Model Fit for Wave 3 (2021)

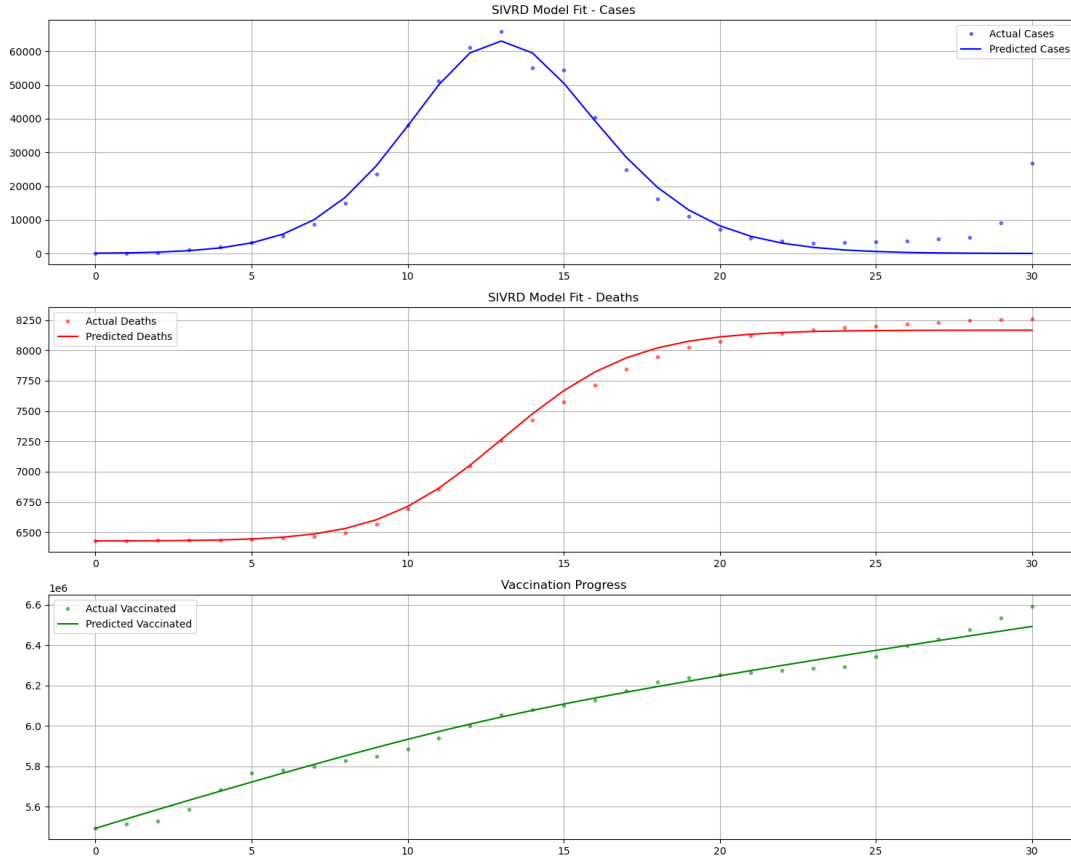


Figure 11: SIVRD Model Fit for Wave 4 (2021)

Comparison of Estimated Parameters with Other Studies

To validate our model's estimated parameters, we compare our results with values reported in a study analyzing COVID-19 spread in Israel [3]. The comparison focuses on key epidemiological parameters: infection rate (β), mortality rate (μ). For this comparison we used average of each wave's parameters for 2020.

Comparison of Weekly Parameters:

Study	β (weekly)	γ (weekly)	μ (weekly)
Our Study (2020, No Vaccine)	17.6455	17.0355	0.01145
Mark Last (2020) [3]	12.537	-	0.091

Table 4: Comparison of Weekly Epidemiological Parameters for Israel

Key Observations:

- Our study's infection rate (β) for 2020 is higher than [3], indicating a more aggressive initial spread in our dataset.
- The mortality rate (μ) in [3] is significantly higher than our estimated values, possibly due to different data sources, estimation methodologies, or early underreporting of cases.

- Our results for 2021 show a substantial decline in both β and μ , aligning with expectations of reduced spread and lower fatality rates following vaccination efforts.

Conclusion

- The SIRD model for 2020 accurately captures infection and mortality trends, but its limitations appear when considering the long-term progression of the pandemic.
- The SIVRD model for 2021 demonstrates the significant impact of vaccination. The estimated vaccination rate (ν) aligns well with real-world data.
- If we had attempted to fit a single model to the entire dataset (without splitting waves), the result would be unrealistic, as it would not account for changing policies, public behavior, and the introduction of vaccines.
- The optimization process successfully estimated β, γ, μ , and ν using a robust least squares approach.

Scenario Analysis

Overview of Predictive Scenarios

To evaluate different possible outcomes of the COVID-19 pandemic in Israel, we simulated multiple scenarios for each wave of infections. These scenarios provide insights into how transmission rates, governmental interventions, and vaccination efforts influenced the pandemic's trajectory.

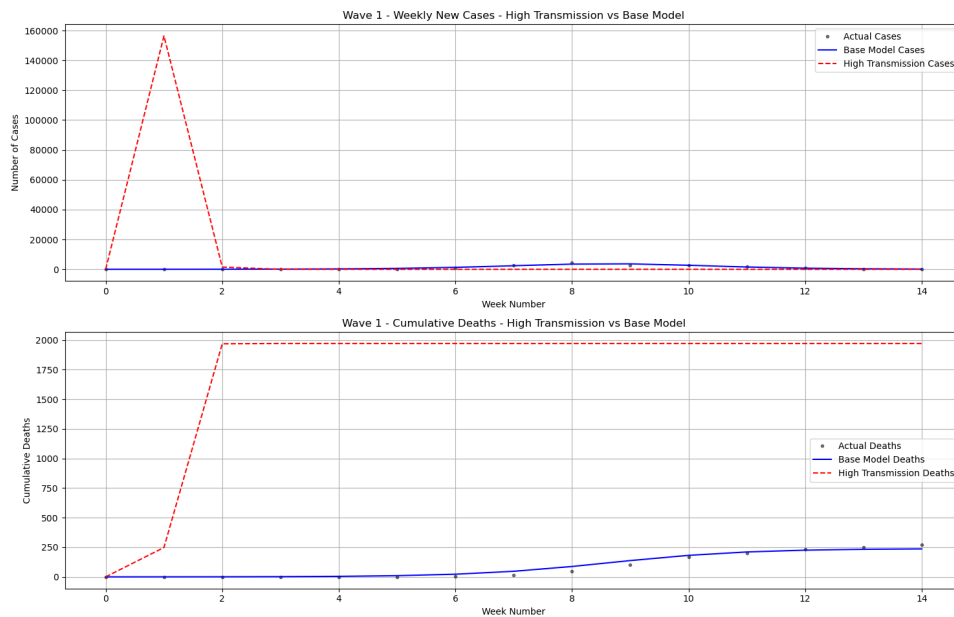
The following scenarios were considered:

- **Base Scenario:** The real-world observed data used as a reference.
- **High Transmission:** Increased transmission rates to reflect reduced restrictions and higher contact rates.
- **Low Transmission:** Lowered transmission rates due to strong public health measures.
- **Optimal Response:** The most effective governmental and public interventions to minimize cases and deaths.
- **No Vaccination (for 2021 waves):** A counterfactual scenario assuming no vaccine rollout.

The results for each wave are summarized below.

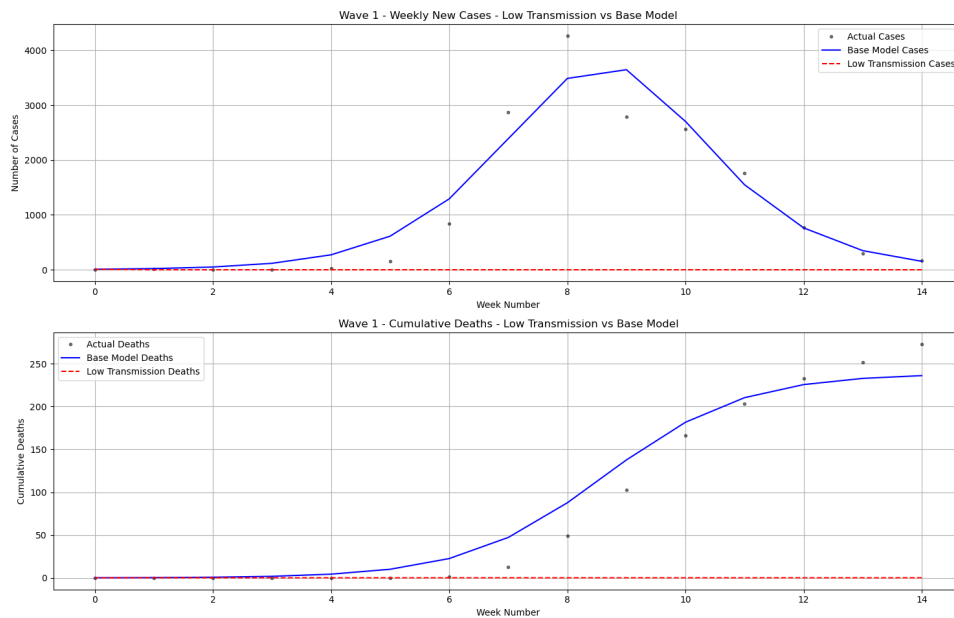
Wave 1: SIRD Model (2020)

SIRD Model - Wave 1 - High Transmission Scenario Analysis



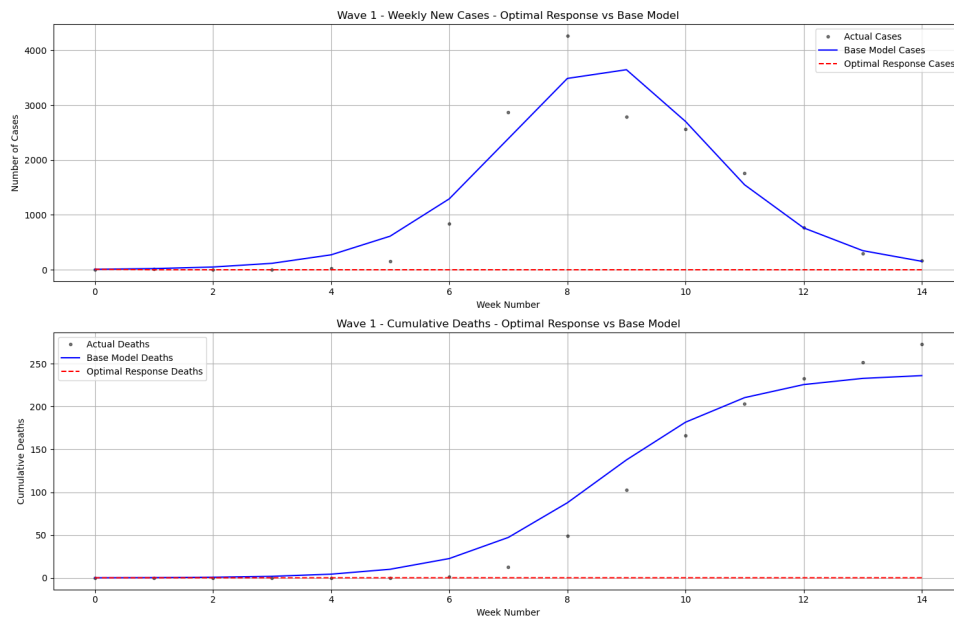
In the high transmission scenario, peak cases surged to 156,391, with deaths increasing to 1,972, demonstrating the severe consequences of uncontrolled spread.

SIRD Model - Wave 1 - Low Transmission Scenario Analysis



The low transmission scenario drastically reduced cases to just 9, with zero deaths, highlighting the effectiveness of strict interventions.

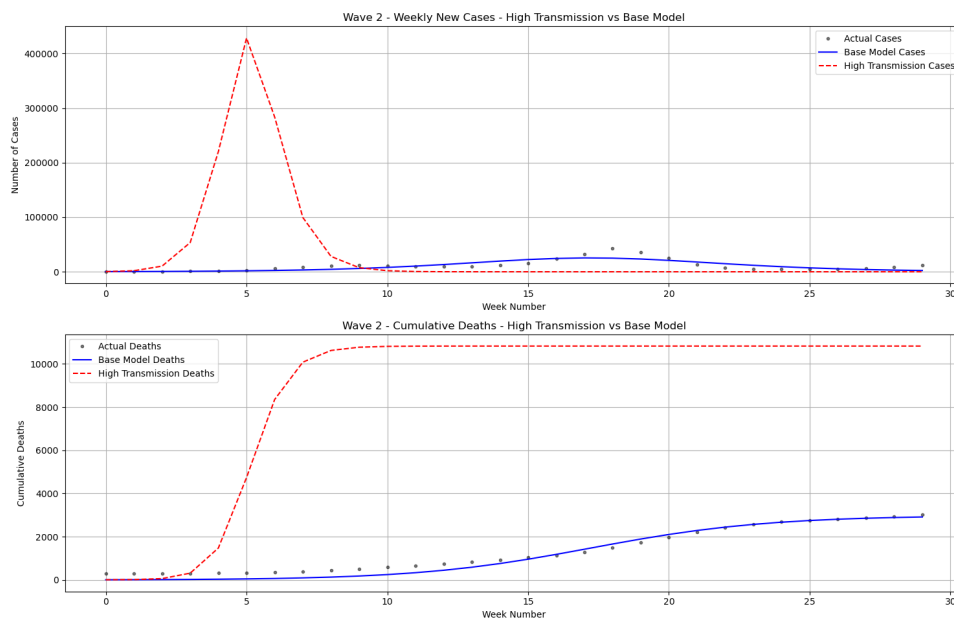
SIRD Model - Wave 1 - Optimal Response Scenario Analysis



The optimal response scenario showed identical results to the low transmission scenario, reinforcing that strict measures can completely curb fatalities.

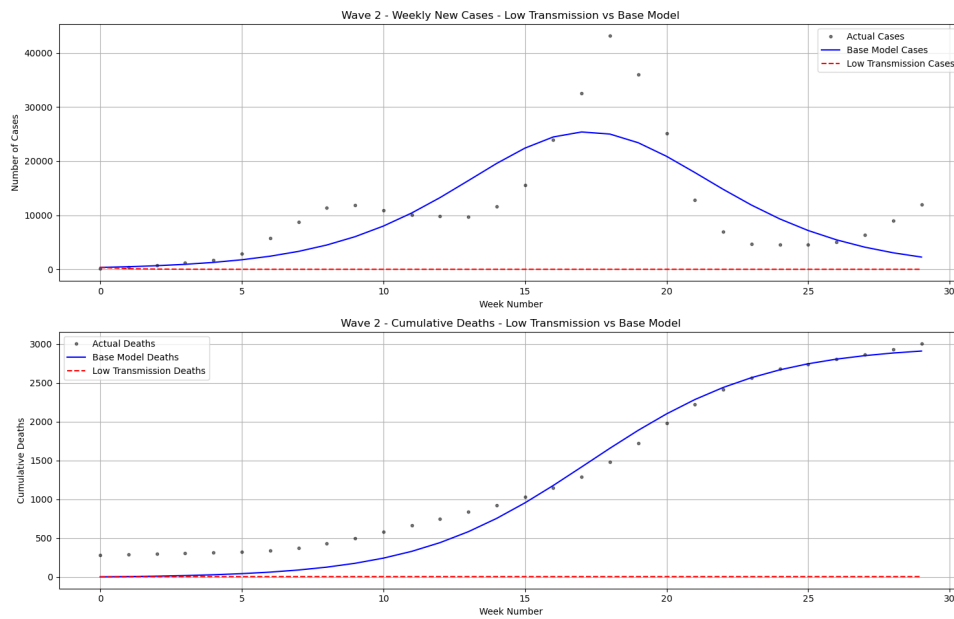
Wave 2: SIRD Model (2020)

SIRD Model - Wave 2 - High Transmission Scenario Analysis



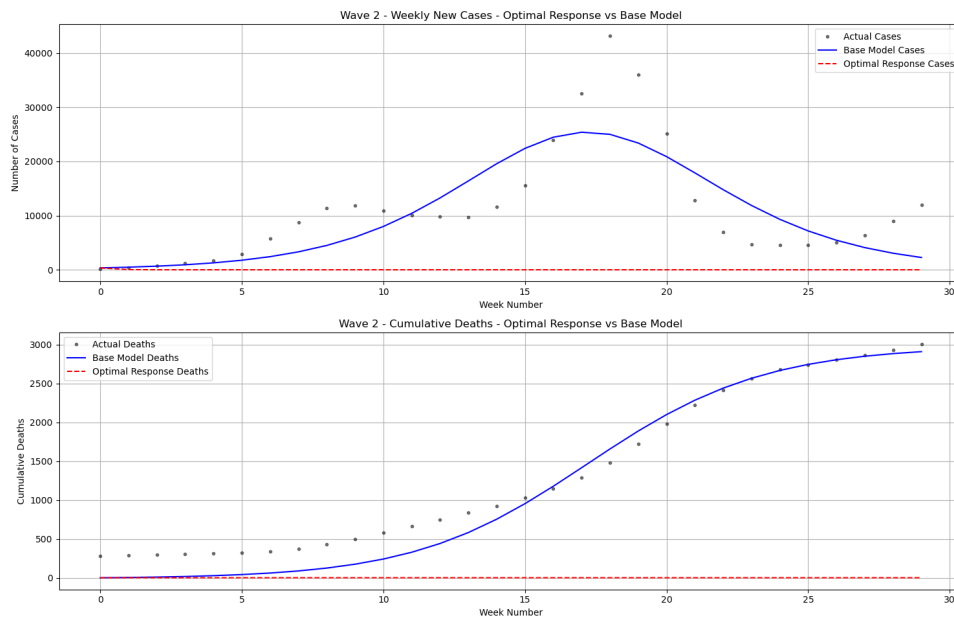
In high transmission conditions, cases peaked at 427,850, with 10,760 deaths, showing how uncontrolled spread could have worsened outcomes.

SIRD Model - Wave 2 - Low Transmission Scenario Analysis



Low transmission significantly mitigated the impact, with only 344 peak cases and 3 deaths.

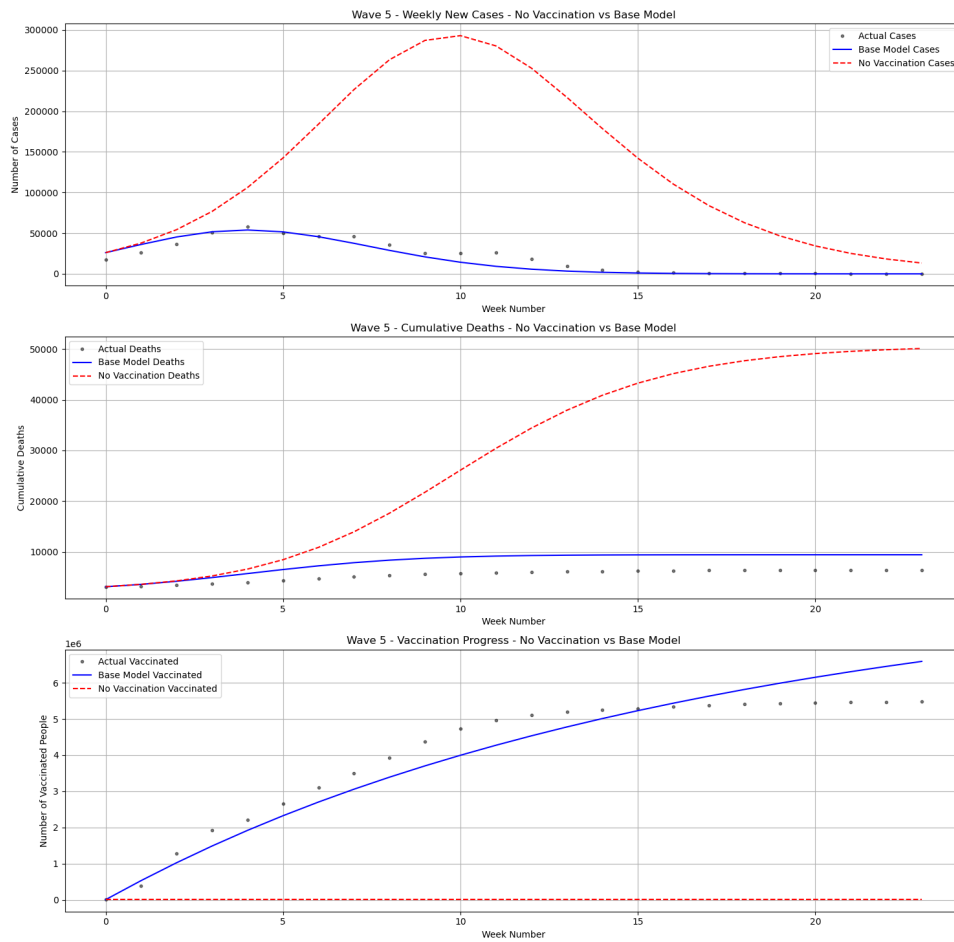
SIRD Model - Wave 2 - Optimal Response Scenario Analysis



The optimal response scenario further reduced deaths to just 1, reinforcing the potential effectiveness of aggressive intervention strategies.

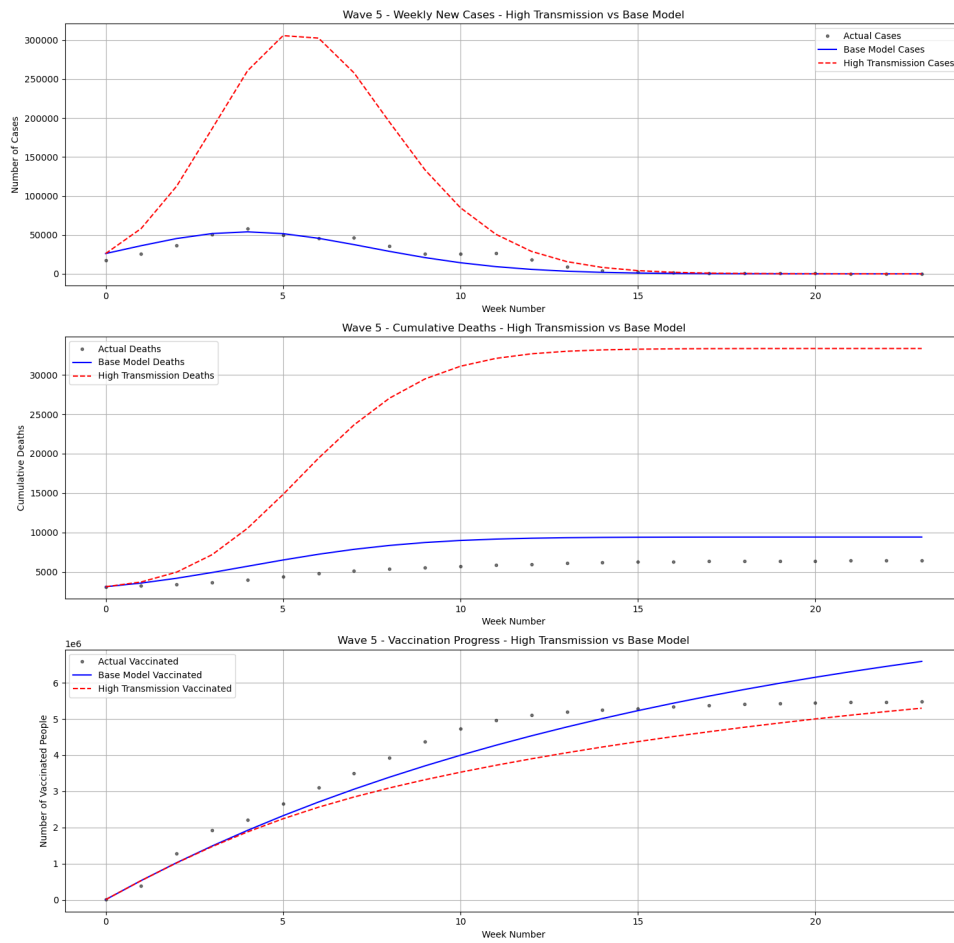
Wave 3: SIVRD Model (2021)

SIVRD Model - Wave 5 - No Vaccination Scenario Analysis



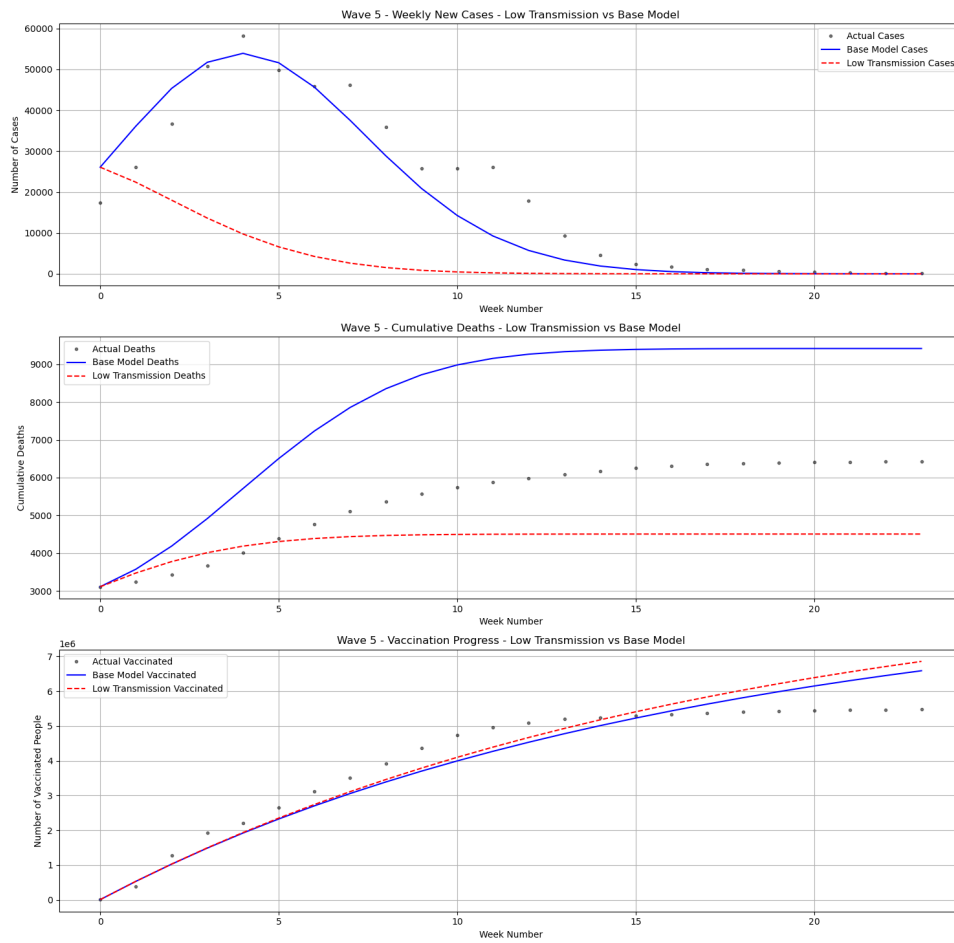
Without vaccination, cases would have skyrocketed to 358,389, and deaths would have reached 33,220, demonstrating the critical role of vaccines.

SIVRD Model - Wave 5 - High Transmission Scenario Analysis



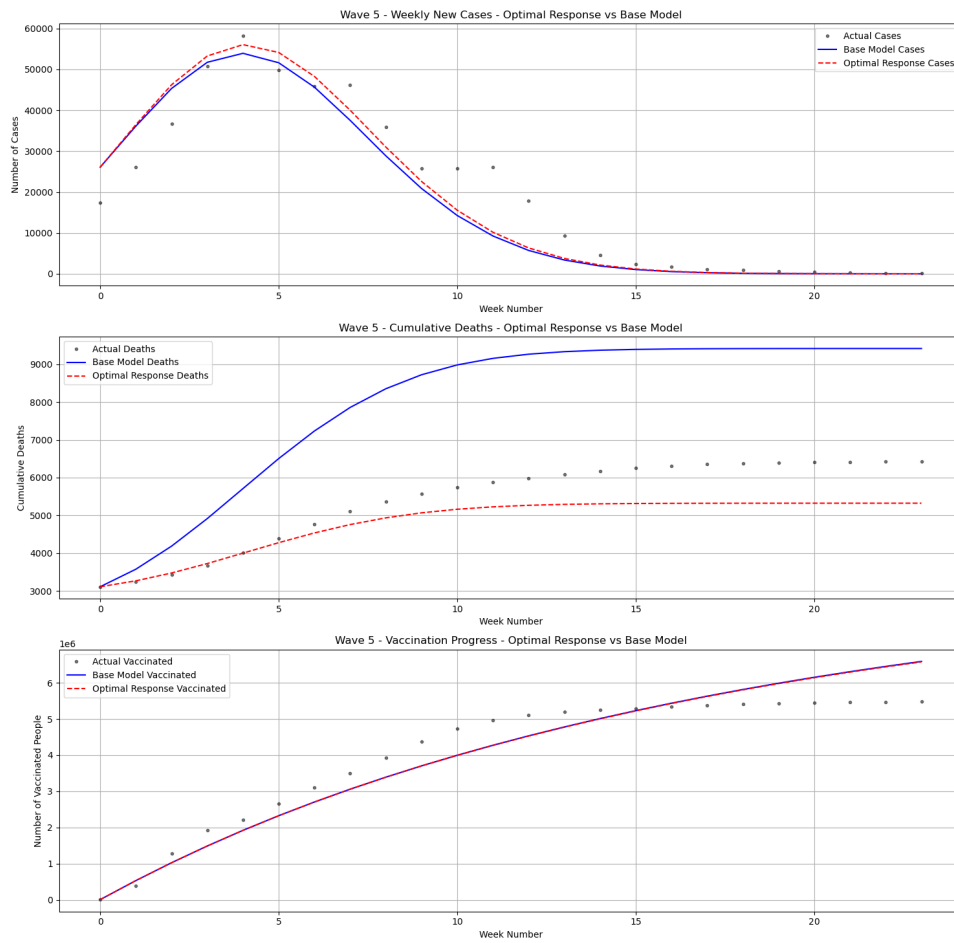
Under high transmission, peak cases rose to 249,836, with 17,445 deaths, further emphasizing the dangers of relaxed measures.

SIVRD Model - Wave 5 - Low Transmission Scenario Analysis



Low transmission reduced cases to 26,083 and deaths to 3,958, proving that non-pharmaceutical interventions still played a role.

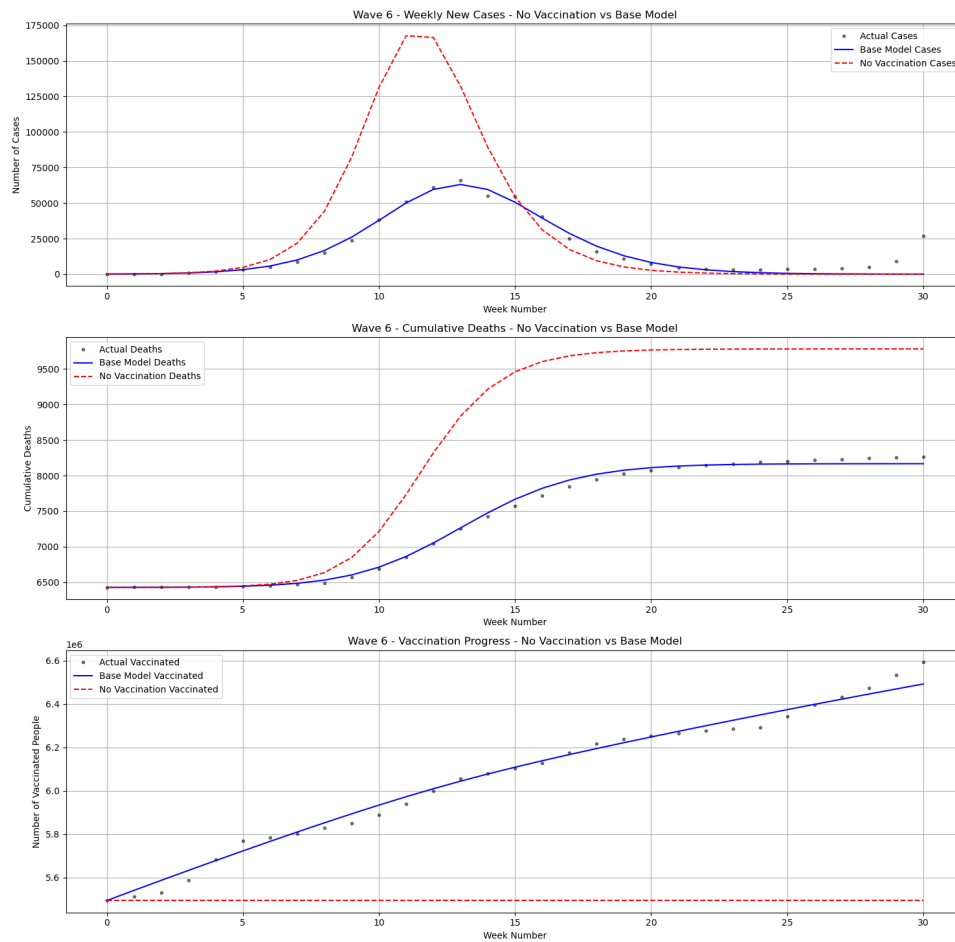
SIVRD Model - Wave 5 - Optimal Response Scenario Analysis



The optimal response scenario showed slightly fewer deaths than the base case, suggesting that while effective, other factors influenced fatality rates.

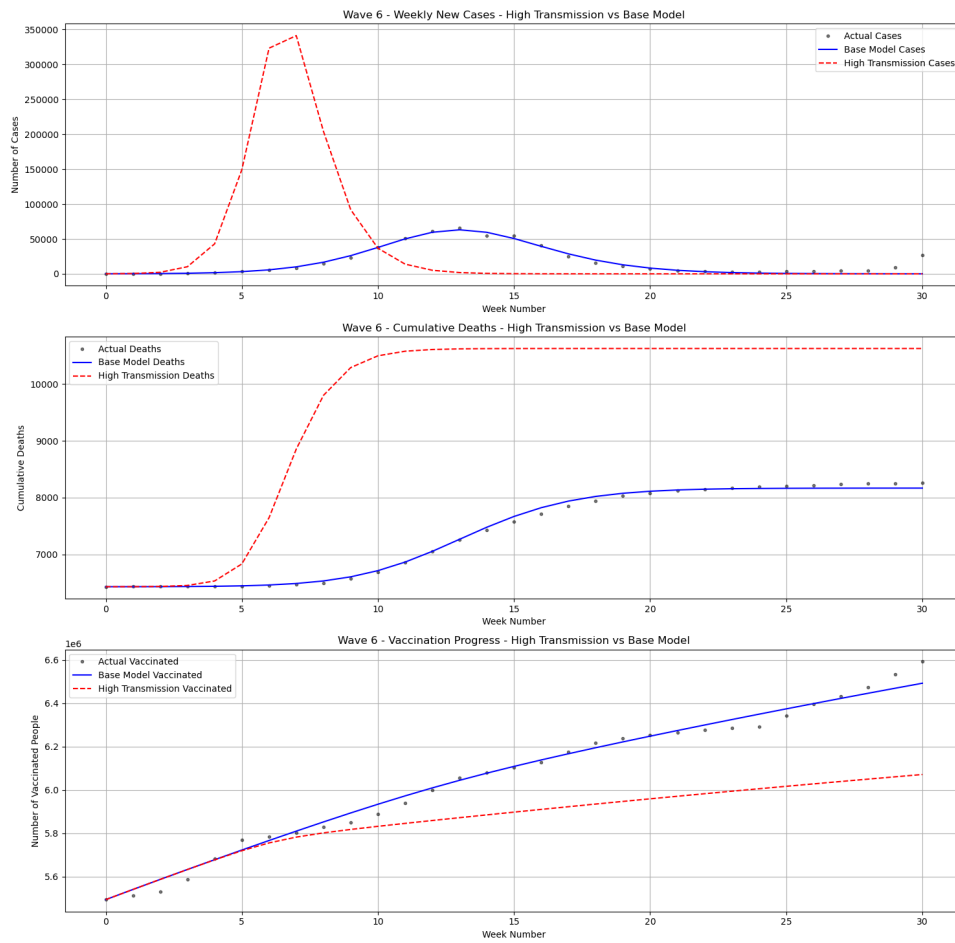
Wave 4: SIVRD Model (2021)

SIVRD Model - Wave 6 - No Vaccination Scenario Analysis



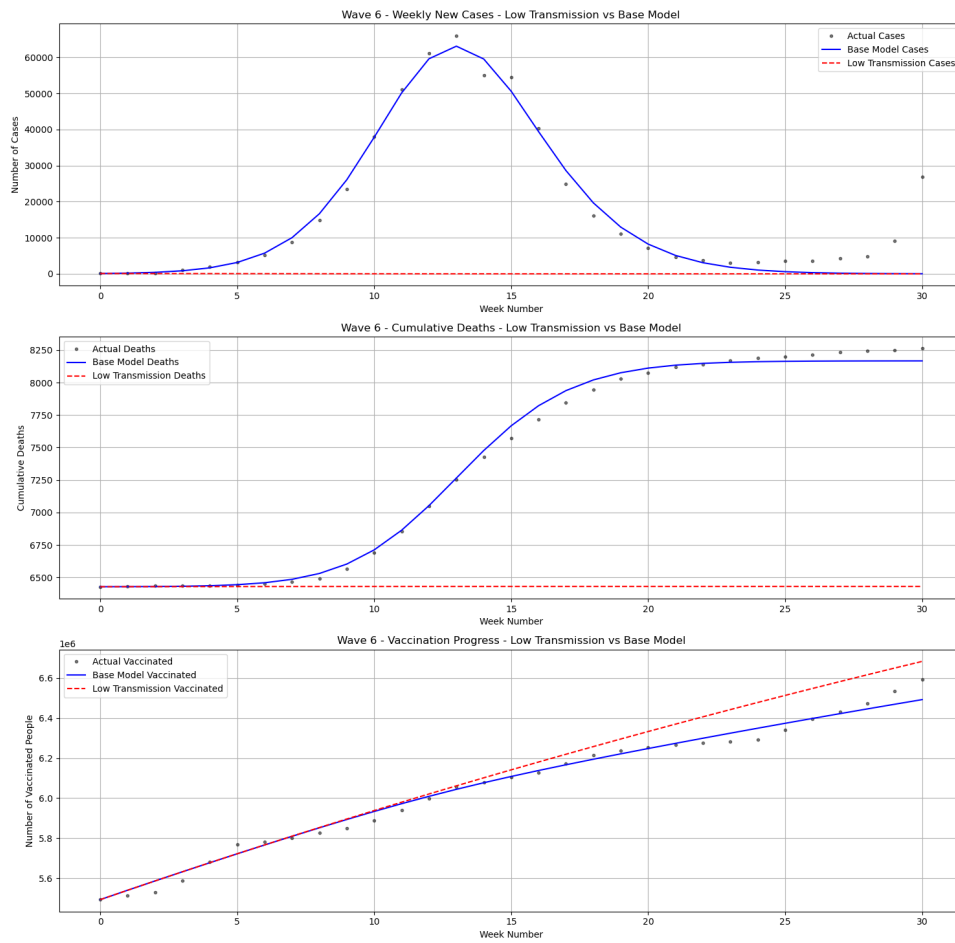
Without vaccination, peak cases increased to 167,684, with deaths rising to 9,780, underscoring the importance of immunization efforts.

SIVRD Model - Wave 6 - High Transmission Scenario Analysis



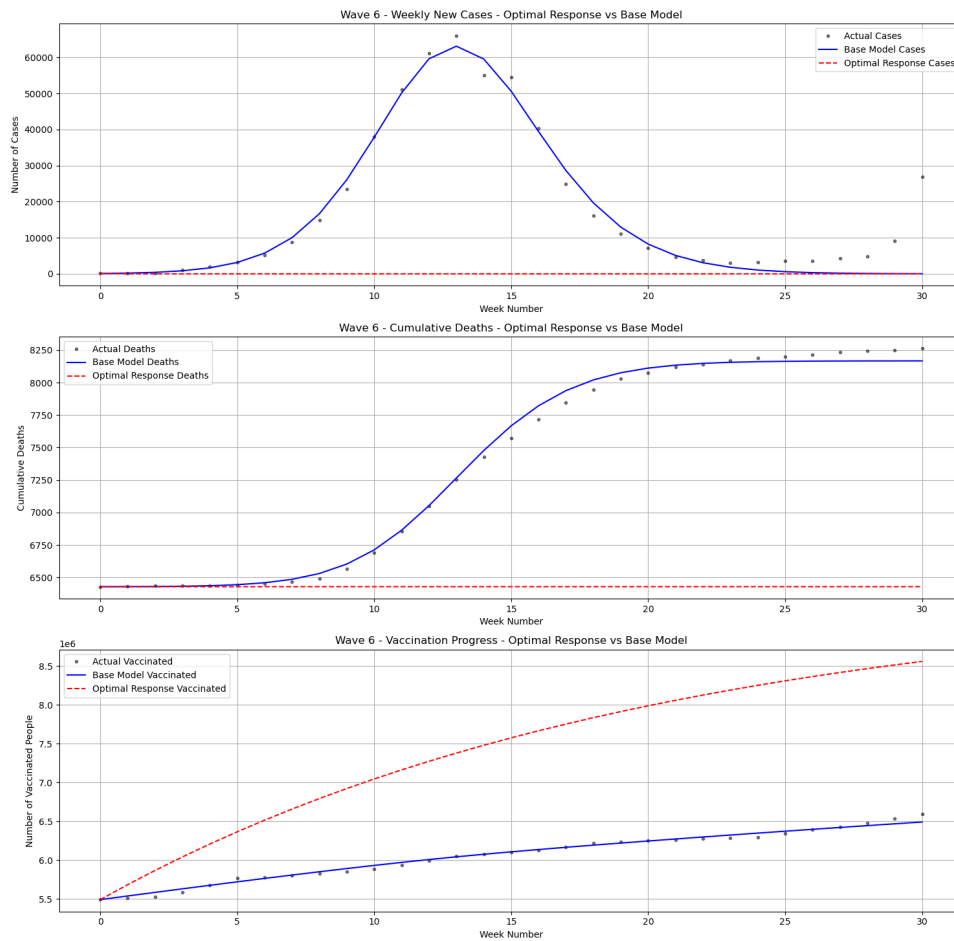
High transmission resulted in 341,300 peak cases and 10,626 deaths, highlighting the compounded risks of both high contact rates and inadequate vaccination.

SIVRD Model - Wave 6 - Low Transmission Scenario Analysis



Low transmission effectively reduced cases and deaths to 92 and 6,431, respectively, showing that containment strategies remained relevant.

SIVRD Model - Wave 6 - Optimal Response Scenario Analysis



The optimal response scenario achieved only a minor improvement over low transmission, indicating that certain structural factors limited further reductions in fatalities.

Key Findings from Scenario Analysis

- High transmission consistently led to a dramatic increase in both cases and deaths.
- Low transmission and optimal response scenarios effectively reduced cases and deaths across all waves.
- The introduction of vaccines in 2021 significantly mitigated fatalities, as seen in Wave 3 and Wave 4.
- In the absence of vaccines, case and death counts would have been substantially higher, reinforcing the importance of vaccination efforts.

This scenario analysis demonstrates the crucial role of public health measures and vaccination in controlling the spread of COVID-19.

Conclusion

This study analyzed the spread of COVID-19 in Israel using statistical and mathematical models, focusing on the impact of government restrictions and vaccination efforts. The key takeaways from our analysis are:

- Government-imposed restrictions had a limited effect in fully controlling the pandemic, as seen in the weak correlation between the stringency index and case numbers in 2020.
- Vaccination played a decisive role in reducing mortality, as shown by the significant drop in the death rate in 2021.
- The SIRD and SIVRD models effectively captured the dynamics of infection waves, validating the importance of vaccination in mitigating severe outcomes.
- Scenario analysis confirmed that high transmission led to extreme case surges, while strong interventions and optimal responses significantly reduced cases and deaths.

Overall, our findings emphasize that while restrictions helped curb transmission, widespread vaccination was the most effective tool in reducing fatalities and managing the pandemic. Future policy decisions should prioritize rapid vaccine deployment alongside targeted non-pharmaceutical interventions to balance public health and economic impact.

References

- [1] Thomas Hale, Noam Angrist, Rafael Goldszmidt, Beatriz Kira, Anna Petherick, Toby Phillips, and Samuel Webster. Oxford covid-19 government response tracker, 2020. Retrieved from https://github.com/OxCGRT/covid-policy-tracker/blob/master/documentation/index_methodology.md.
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